Chutong Yang

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Research Focus

My research focuses on learning theory and algorithms for reliable machine learning, with an emphasis on calibration and optimization. My recent work explores LLM reasoning and multi-agent collaboration, including AI agents for automated research, with additional interests in parallel sampling for diffusion and flow models.

Education

The University of Texas at Austin, Ph.D. in Computer Science — Aug 2023 – Present
- Recipient, **Amazon AI PhD Fellowship**

Stanford University, M.S. in Computer Science — Sep 2021 – Jun 2023
- Full financial support for two years.

University of California, San Diego, B.S. in Computer Science and Mathematics — Sep 2016 – Jun 2020

Experience

Amazon Web Services (AWS), Applied Scientist Intern — Summer 2026
- Built TCSAlgBench, a benchmark of sandboxed, research-level theoretical computer science challenges for evaluating LLMs and agentic systems.
- Designed and evaluated agent harnesses, including plan-and-execute and Monte Carlo tree search (MCTS), finding that agentic planning performed best among the evaluated approaches.

The University of Texas at Austin, Research Assistant — Aug 2023 – Present
- Developed optimal online recalibration algorithms, with simultaneous calibration and squared-loss improvement guarantees for discretized forecasts under adversarial data sequences.
- Develop methods for reliable LLM reasoning through uncertainty-guided multi-agent deliberation and RL-based confidence calibration.

Stanford University, Research Assistant — 2022 – 2023
- Researched algorithmic fairness with Prof. Omer Reingold, focusing on multicalibration.
- Developed optimization algorithms with Prof. Aaron Sidford using interior-point methods.

Skills

Programming: Python, C++, Java, MATLAB **ML:** PyTorch

Research Software

Algo2Lean – Independent collaboration with Lunjia Hu
- Contributed to a language and compiler that translates Python-like algorithms into Lean 4 definitions, supporting formal proofs of correctness and running-time bounds.
- Makes formalized algorithms and their output guarantees easier to read, understand, and review.

Publications

(Authors are listed alphabetically unless otherwise indicated by *.)

13. TCSAlgBench: Benchmarking Automated Proving for Research-Level Theoretical Computer Science — Preprint, 2026

Chutong Yang*, Xiyuan Zhang, Yu Huang, Boran Han, Soonho Kong, Shuai Zhang, Vihang Prakash Patil, Zhen Han, Michael Bohlke-Schneider, Bernie Wang

http://arxiv.org/abs/2609.35606

12. Gap-Free Streaming PCA Beyond Rank-One Updates: Near-Optimal Rates and Applications to Differential Privacy Preprint, 2026

Anming Gu, Syamantak Kumar, Kevin Tian, ***Chutong Yang***

https://arxiv.org/abs/2609.26508

11. Optimal Recalibration of an Online Predictor **NeurIPS** 2026 (spotlight)

Lunjia Hu, Kevin Tian, ***Chutong Yang***

https://arxiv.org/abs/2607.19689

10. Simultaneous Blackwell Approachability and Applications to Multiclass Omniprediction **COLT** 2026

Lunjia Hu, Kevin Tian, ***Chutong Yang***

https://arxiv.org/abs/2602.17577

9. Spherical Leech Quantization for Visual Tokenization and Generation **CVPR** 2026 (Highlight)

Yue Zhao*, Hanwen Jiang, Zhenlin Xu, ***Chutong Yang***, Ehsan Adeli, Philipp Krähenbühl

https://arxiv.org/abs/2512.14697

8. Proportionality from Low-Dimensional Approval Data **AAMAS** 2026

Zhiyi Huang, Gregory Kehne, ***Chutong Yang***

Extended abstract: https://doi.org/10.65109/YHKI7695

7. Private Geometric Median in Nearly-Linear Time **NeurIPS** 2025

Syamantak Kumar, Daogao Liu, Kevin Tian, ***Chutong Yang***

TPDP 2025 Workshop

https://arxiv.org/abs/2505.20189

6. Omnipredicting Single-Index Models with Multi-Index Models **STOC** 2025

Lunjia Hu, Kevin Tian, ***Chutong Yang***

https://arxiv.org/abs/2411.13083

5. Testing Calibration in Nearly-Linear Time **NeurIPS** 2024

Lunjia Hu, Arun Jambulapati, Kevin Tian, ***Chutong Yang***

https://arxiv.org/abs/2402.13187

4. Omnipredictors for Constrained Optimization **ICML** 2023

Lunjia Hu, Inbal Livni-Navon, Omer Reingold, ***Chutong Yang***

https://arxiv.org/abs/2209.07463

3. Active Learning Polynomial Threshold Functions **NeurIPS** 2022

Omri Ben-Eliezer, Max Hopkins, ***Chutong Yang***, Hantao Yu

https://arxiv.org/abs/2201.09433

2. A Fast Exact Algorithm for Deployment of Sensor Nodes for Internet of Things **ISF** 2020

Qinghua Zheng*, ***Chutong Yang***, Haijun Yang, Jianhe Zhou

Information Systems Frontiers: https://link.springer.com/article/10.1007/s10796-018-9890-3

1. Detailed placement for IR drop mitigation by power staple insertion in sub-10nm VLSI **DATE** 2019

Sun ik Heo, Andrew B Kahng, Minsoo Kim, Lutong Wang, ***Chutong Yang***

https://ieeexplore.ieee.org/abstract/document/8715096

Professional Service

Conference reviewer: NeurIPS 2024, ICLR 2025, ICML 2025, ITCS 2026, NeurIPS 2026, ICLR 2027